Anvitha Reddy

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Summary

AI Engineer specializing in production LLM systems—retrieval-augmented generation, natural-language-to-SQL, and multi-model LLM orchestration—with a reliability-first approach spanning validation, circuit breakers, and CI/CD. At AMD, shipped **ZenBot** to deliver **up to 120× faster** product-health analysis with near-zero hallucination (parameterized SQL + AST-level validation), recognized with a Software Poster at the AMD Asia Technical Conference (AATC) 2026. Previously engineered a real-time, **50+ language** speech engine for a client-funded AI product.

Experience

Advanced Micro Devices (AMD) **Aug 2025 – Present**
AI Engineer, ZenBot Enterprise AI Agent *Hyderabad, India*

- Designed and shipped **ZenBot**, an enterprise agent answering plain-language questions across four previously disconnected enterprise systems (JIRA, JAMA, test automation, and report repositories), reducing a recurring **2–4 hour manual analysis to under 2 minutes** for non-technical stakeholders.
- Architected a two-model LLM cascade: a lightweight intent classifier that routes to a SQL-generation model, enabling zero-SQL data access while lowering inference cost and latency.
- Reduced hallucination to near-zero by defaulting to parameterized SQL for known queries and validating every model-generated query at the AST level (sqlglot), backed by a schema-grounded auto-repair loop.
- Integrated semantic retrieval (RAG) over the enterprise report corpus into the unified query layer, with report-type disambiguation across five categories, exposed through **12+ MCP tool endpoints**.
- Hardened the on-premises deployment with circuit breakers, TTL caching, and graceful degradation (no cascading failures at peak load), and built a CI/CD pipeline (ruff, mypy, unit and integration tests) enforcing a 70% coverage gate with config-driven onboarding of new domains.

CornerstoneX **Nov 2024 – Aug 2025**
AI Engineer *Remote*

- Built and shipped the **real-time multilingual speech engine** behind a **client-funded** agentic-AI metahuman, **deploying 50+ languages to production** so users speak in their own language and receive spoken, avatar-delivered answers in near real time.
- Architected an English-pivot translation layer composing **50+ Helsinki-NLP OPUS-MT bilingual models with an M2M100 fallback** for uncovered language pairs, GPU-accelerated on CUDA to sustain live, real-time conversation.
- Engineered an LLM-driven voice assistant that maps natural-language commands to structured browser and desktop actions (Selenium, Playwright), enabling hands-free multi-step navigation for users with mobility challenges.
- Trained a CNN-based ASL gesture-recognition model for real-time sign-language translation, delivered for a **U.S. Department of Homeland Security** client in public-transit settings.
- Contributed to a platform selected for the **NVIDIA Inception Program** and presented at **NVIDIA GTC 2025**.

Education

Christ University **Aug 2021 – May 2025**
B.Tech in Computer Science (Data Science) *Bengaluru, India GPA: 3.5/4.0*

Technical Skills

Languages: Python, SQL
LLM & GenAI: Retrieval-Augmented Generation (RAG), Natural-Language-to-SQL, Agentic AI & Multi-Agent Systems, LLM Orchestration & Multi-Model Routing, Tool Use / Function Calling, Prompt Engineering, Embeddings & Semantic Search
NLP, Speech & Vision: Machine Translation (OPUS-MT, M2M100), Speech-to-Text / Text-to-Speech, Multilingual Pipelines, Computer Vision (CNN)
Frameworks & Libraries: LangChain, LlamaIndex, Hugging Face, Whisper, sqlglot, FastAPI, Selenium, Playwright, ElevenLabs, Ollama
APIs, Vector DBs & Platforms: OpenAI API, Anthropic Claude API, MCP (Model Context Protocol), Pinecone, Weaviate, FAISS, Snowflake, CUDA, Docker, Git, Linux, n8n
Reliability & Delivery: AST Validation, Auto-Repair Loops, Circuit Breakers, TTL Caching, Graceful Degradation, CI/CD (ruff, mypy), Unit & Integration Testing